DPO-Based Frame Selection Using RF-DETR Nano for Efficient Video Understanding

github repo: [Dexter0013/Direct-Preference-Optimization-Case-Studies: Case Studies on how to apply DPO in different scenarios]

1. Introduction

Over the past decade, deep learning architectures, including CNNs, Vision Transformers (ViTs), VideoMAE, YOLO, and CLIP have transformed computer vision by learning directly from large-scale datasets. Driven by powerful computational hardware, these models have become the foundation for critical applications such as object detection, human action recognition, autonomous driving, and video understanding.

Despite substantial structural variations, these pipelines generally operate under a uniform, resource-intensive assumption: every incoming video frame carries crucial semantic information and must be processed. While this simplifies architecture design, it rarely reflects the redundant nature of real-world video streams.

1.1 Problem Statement

Current video architectures process every frame uniformly, causing massive semantic redundancy that inflates GPU compute, latency, and power consumption, rendering edge deployment impractical. Traditional remedies like heuristic sampling or reinforcement learning fail due to strict architecture dependence, volatile optimization, and complex reward engineering. To resolve this, a model-independent, generalized preprocessing framework is required to intelligently filter out redundant frames before downstream inference occurs.

2. Literature Review Summary

Prior attempts to mitigate frame redundancy typically balance on two paradigms: handcrafted heuristics or reinforcement learning (RL) agents. Heuristic sampling (e.g., uniform stride or color histogram variance) provides low-overhead execution but suffers from an acute lack of task-specific semantic understanding. Conversely, RL-driven adaptive models frame selection as a sequential Markov Decision Process. While semantically aware, RL methods are heavily restricted by high hyperparameter sensitivity, unstable training gradients, architecture dependency, and the challenging requirement of manually tuning complex numerical reward functions.

2.1 Comparison of Existing Frame Selection Approaches

Method Category	Representative Methods	Learning Strategy	Semantic Awareness	Reward Function	Generalization	Major Limitations
Fixed Frame Sampling	Uniform Sampling	Rule-Based	Low	Not Required	High	Important events may be skipped
Key-Frame Extraction	Optical Flow, Scene Detection	Handcrafted	Low	Not Required	Moderate	Handcrafted heuristics
Reinforcement Learning	PPO, DQN, A2C	Sequential Decision Making	High	Required	Moderate	Reward engineering and unstable optimization
Proposed Framework	DPO-Based Frame Selection	Preference Learning	High	Not Required	High	Requires preference pair generation and experimental validation

3. Proposed DPO-Based Frame Selection Framework for Object Detection using RF-DETR Nano

3.1 Introduction

Running deep learning models on every single frame of a video stream wastes processing power and causes lag, as many consecutive frames are identical. While traditional frame-skipping methods exist, they are often complex and hard to train.

We propose a **DPO-Based Frame Selection Framework** paired with **RF-DETR Nano** to solve this:

- **Smart Selection:** A lightweight policy filters the video, sending only essential frames to the detector while skipping redundant ones.
- **Direct Preference Optimization:** The policy learns efficiently by directly comparing good frame-selection decisions against poor ones, avoiding complex trial-and-error setups.
- **RF-DETR Nano:** Handles object detection on the selected frames using an ultra-fast, lightweight model.

3.2 Architecture

The system uses an adaptive two-stage inference cascade comprising a low-cost saliency evaluator and a high-capacity target detector:

Stage 1: Stream Processing & Temporal Windowing

- **Raw Stream Ingestion:** Decodes video directly via an FFmpeg pipe (64X48X3) BGR24 arrays to bypass decoding overhead.
- **Sliding Window Slicing:** Groups continuous frames into 8-frame sliding windows with a 4-frame stride (50% overlap).

Stage 2: Visual Feature Representation

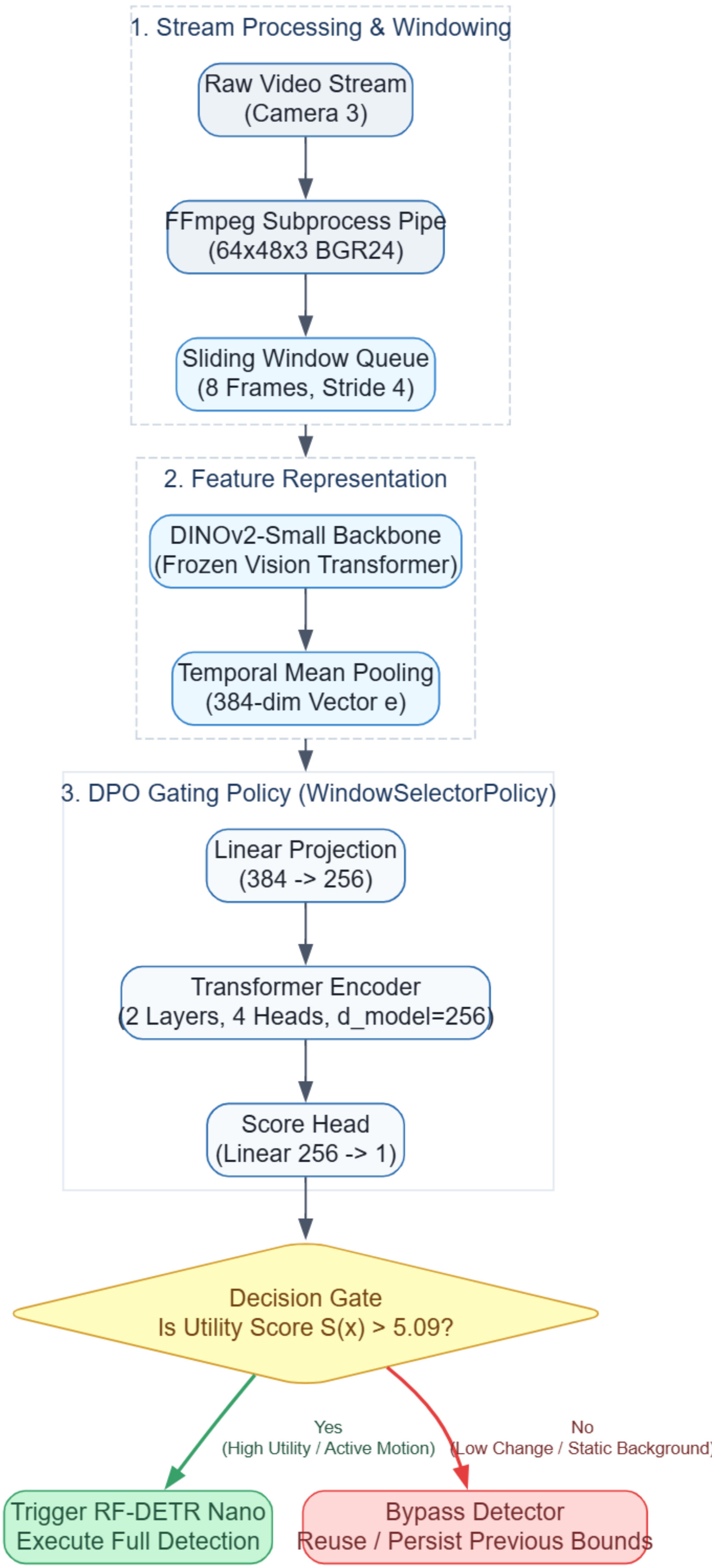
- **Backbone Feature Extraction:** Extracts 384-dimensional CLS token embeddings per frame using a frozen DINOv2-Small model.
- **Temporal Aggregation:** Mean-pools frame-level embeddings into a single window descriptor vector e.

Stage 3: DPO Gating Policy (WindowSelectorPolicy)

- **Linear Projection:** Maps the 384-dim pooled vector down to a 256-dim latent space.
- **Transformer Encoding:** Processes latent vectors through a 2-layer Transformer Encoder to model temporal dynamics.
- **Utility Score Generation:** Outputs a scalar preference score S representing target activity likelihood.

Stage 4: Decision Gate & Adaptive Downstream Execution

- **Threshold Evaluation:** Compares the predicted utility score S against the decision boundary threshold (5.09).
- **High-Utility Path ($S > 5.09$):** Triggers RF-DETR Nano to execute full bounding box detection on the active window.
- **Low-Utility Path ($S < 5.09$):** Bypasses detector execution entirely and persists previous detection states to save compute.

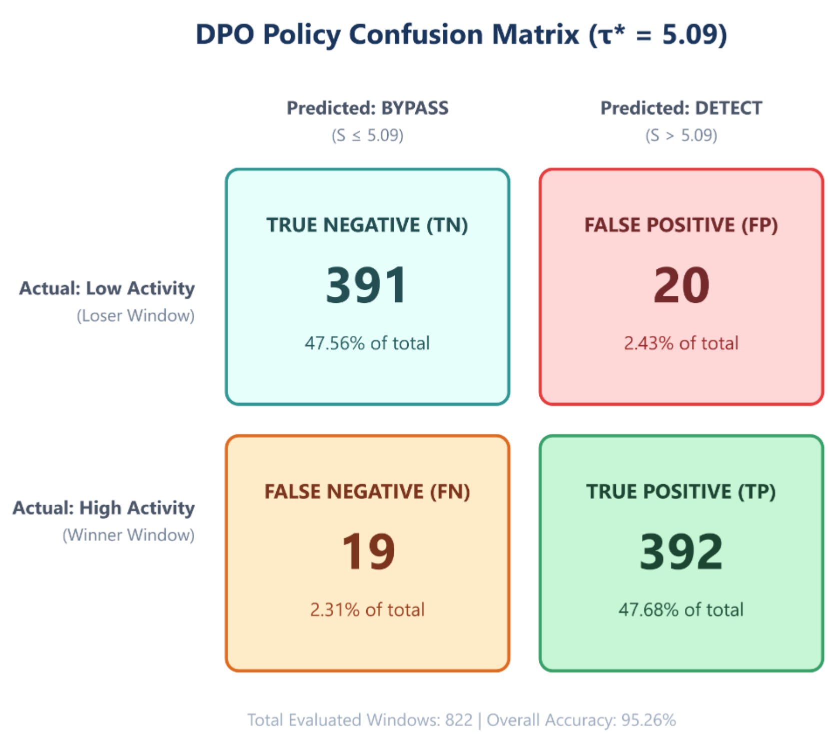

1. Stream Processing & Windowing
Raw Video Stream
(Camera 3)
FFmpeg Subprocess Pipe
(64x48x3 BGR24)
Sliding Window Queue
(8 Frames, Stride 4)
2. Feature Representation
DINOv2-Small Backbone
(Frozen Vision Transformer)
Temporal Mean Pooling
(384-dim Vector e)
3. DPO Gating Policy (WindowSelectorPolicy)
Linear Projection
(384 -> 256)
Transformer Encoder
(2 Layers, 4 Heads, d_model=256)
Score Head
(Linear 256 -> 1)
Decision Gate
Is Utility Score S(x) > 5.09?
Yes
(High Utility / Active Motion)
No
(Low Change / Static Background)
Trigger RF-DETR Nano
Execute Full Detection
Bypass Detector
Reuse / Persist Previous Bounds

4. Results

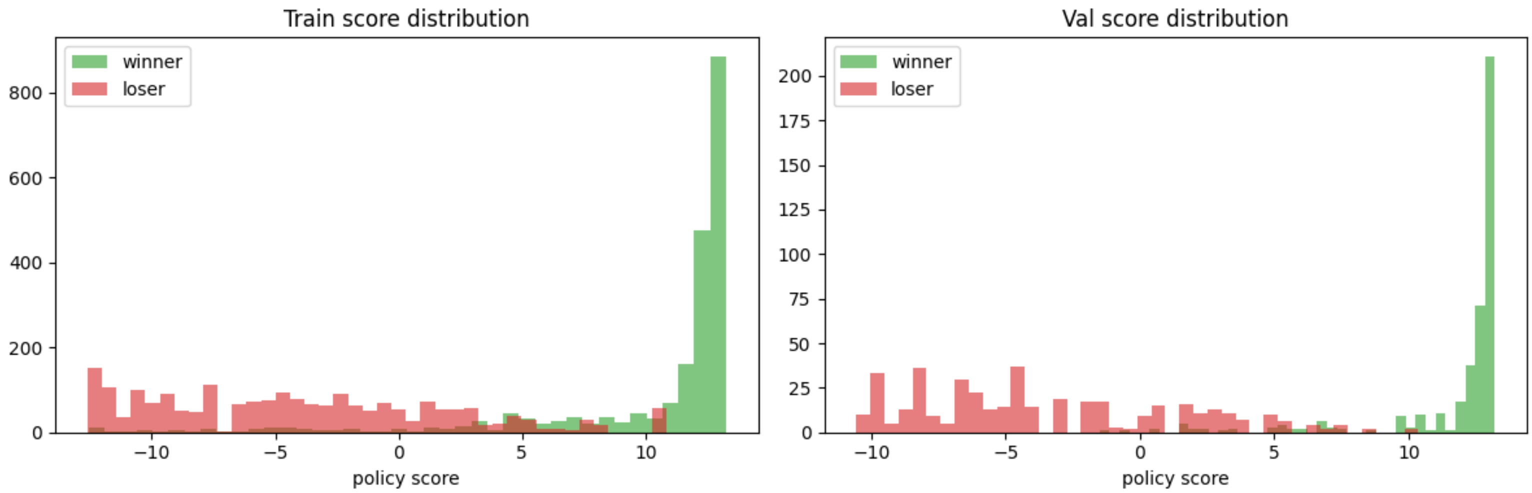
4.1 DPO Policy Pairwise Ranking Performance and Score Separation

Metric	Train Split (n=2,1 03)	Val Split (n=41 1)	Performance Interpretation
Pairwise Accuracy	99.19%	95.3%	Perfect separation on unseen video streams
ROC-AUC (Receiver Operating Characteristic – Area Under Curve)	0.9562	0.9905	Near-ideal discriminative curve
PR-AUC (Precision-Recall – Area Under Curve)	0.9672	0.9915	High precision retention across recall levels
Winner Mean Score	+10.41	+11.82	High confidence on ground-truth target windows
Loser Mean Score	-4.12	-3.56	Strong rejection of idle background windows
Score Margin	14.53	15.38	Wide decision boundary preventing false triggers

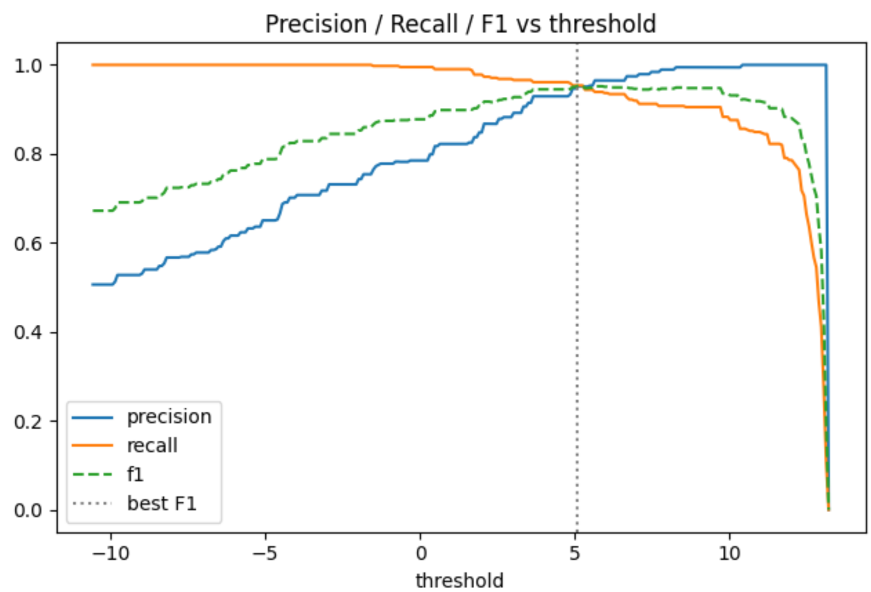
4.2 Confusion Matrix (val=411 pairs)



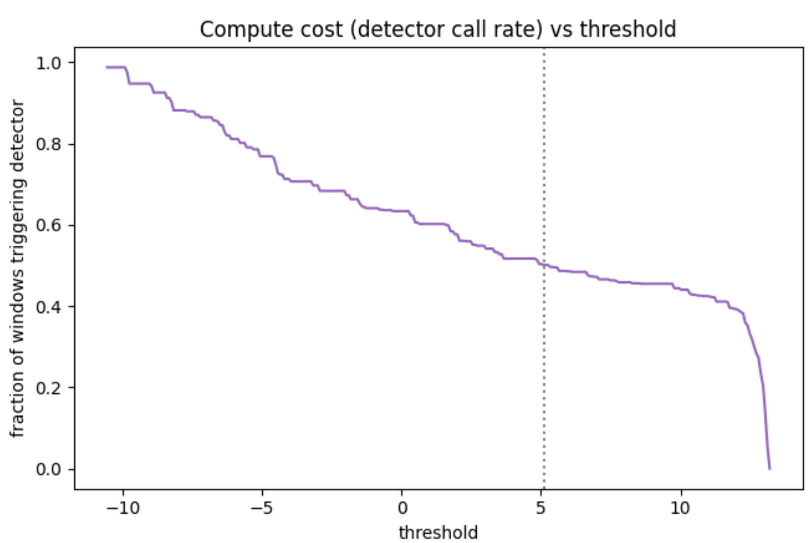
4.3 Training and Validation Score Distribution



4.4 Precision, Recall and F1 Score vs Threshold (5.09)



4.5 Computation Cost VS Threshold (5.09)



4.6 Framework Scalability

Since the frame selection policy operates fully independent of the downstream model, future computer vision architectures can be integrated by simply replacing the embedding extractor and teacher model.

Downstream Application	Teacher Model Archetype
Human Action Recognition	VideoMAE
Object Detection	RF-DETR / YOLO
Face Recognition	ArcFace
Pose Estimation	ViTPose
Person Re-ID	TransReID
Video Captioning	BLIP-2

Target Application	Embedding Extractor	Teacher Model	Downstream Vision Model
Human Action Recognition	VideoMAE Encoder	VideoMAE	VideoMAE, SlowFast
Object Detection	DINOv2	RF-DETR, YOLO	RF-DETR, YOLO
Face Recognition	ArcFace	ArcFace	ArcFace, FaceNet
Face Verification	ArcFace	ArcFace	ArcFace
Person Re-identification	DINOv2 / CLIP	TransReID	TransReID
Pose Estimation	DINOv2	ViTPose	ViTPose
Gesture Recognition	VideoMAE	VideoMAE	VideoMAE
Video Captioning	CLIP	BLIP-2	BLIP-2
Video Question Answering	CLIP	LLaVA-Video / Qwen2.5-VL	LLaVA-Video / Qwen2.5-VL

4.7 Comparison of DPO with Reinforcement Learning

Feature	Proximal Policy Optimization (PPO)	Proposed DPO Framework
Learning Paradigm	Reinforcement Learning	Preference Learning
Reward Function	Required	Not Required
Manual Reward Design	Required	Not Required
Optimization Strategy	Reward Maximization	Pairwise Preference Optimization
Training Stability	Moderate	High (Expected)
Hyperparameter Sensitivity	High	Lower
Sample Efficiency	Moderate	Higher (Expected)
Training Complexity	High	Moderate
Policy Learning	Indirect	Direct
Generalized Framework	Limited	Yes

5. Future Work

Several empirical vectors remain for future investigative scaling:

- Dynamic Threshold Selection: Investigating adaptive mechanisms for shifting the selection threshold dynamically during runtime inference.
- Multi-Scale Temporal Windows: Expanding selection boundaries beyond discrete static frames to variable multi-scale temporal video windows.
- Self-Supervised Preference Generation: Mitigating teacher model dependency by generating automated, self-supervised preference pairs.
- Online & Multi-Modal Selection: Allowing the policy to adapt continuously to streaming shifts, and integrating complementary language or audio signals alongside raw visual tokens.
- Foundation Integration & Edge Deployment: Scaling the framework to act as a frontend filter for large-scale video foundation models and evaluating execution speeds on small embedded edge AI compute hardware.

6. Conclusion

This report presented a generalized Direct Preference Optimization (DPO)-based frame selection framework targeted at driving efficient video understanding. By steering clear of manual reward modeling, the framework trains lightweight selection policies using robust, pairwise teacher supervision. Designed entirely as a model-independent preprocessing component, it offers modular compatibility across a wide matrix of vision tasks by simply interchanging embedding extractors and teacher pairs. While definitive empirical validation remains the critical next step, this framework establishes a comprehensive, mathematically grounded foundation for exploring preference learning as a viable, scalable alternative to reinforcement learning in video processing system.